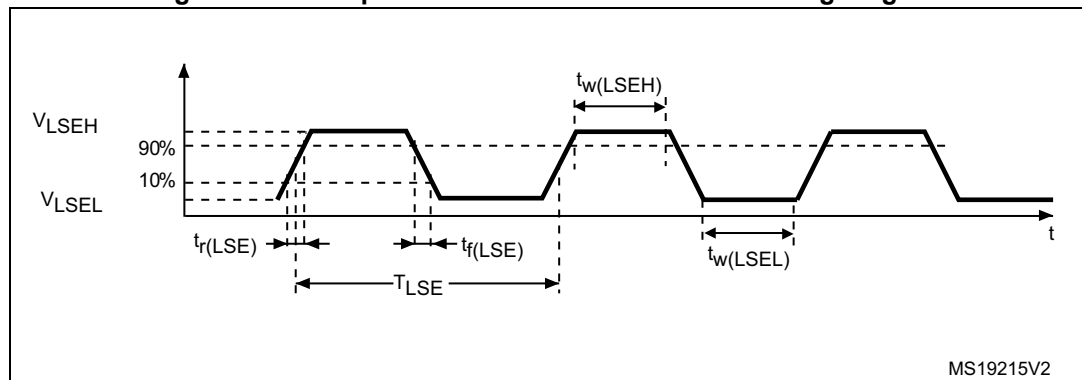


Table 37. Low-speed external user clock characteristics (continued)

Symbol	Parameter ⁽¹⁾	Conditions	Min	Typ	Max	Unit
V_{LSEH}	OSC32_IN input pin high level voltage	-	$0.7 \times V_{DDIO1}$	-	V_{DDIO1}	V
V_{LSEL}	OSC32_IN input pin low level voltage	-	V_{SS}	-	$0.3 \times V_{DDIO1}$	
$t_{w(LSEH)}/t_{w(LSEL)}$	OSC32_IN high or low time	-	250	-	-	ns

1. Specified by design. Not tested in production.

Figure 13. Low-speed external clock source AC timing diagram



High-speed external clock generated from a crystal/ceramic resonator

The high-speed external (HSE) clock can be supplied with a 4 to 48 MHz crystal/ceramic resonator oscillator. All the information given in this paragraph are based on design simulation results obtained with typical external components specified in [Table 38](#). In the application, the resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics (frequency, package, accuracy).

Table 38. HSE oscillator characteristics

Symbol	Parameter ⁽¹⁾	Conditions ⁽²⁾	Min	Typ	Max	Unit
f_{OSC_IN}	Oscillator frequency	-	4	-	48	MHz
R_F	Feedback resistor	-	-	200	-	k Ω